

ZAPATA COUNTY AP, TX, USA

WMO: 720584

Lat: 26.967N

Lon: 99.250W

Elev: 424

StdP: 14.47

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 00181

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	32.2	36.3	13.6	11.2	56.5	18.0	13.8	54.0	22.4	67.0	19.8	63.6	2.7	350	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	23.9	104.4	71.8	102.5	72.3	100.6	72.5	78.0	87.4	77.3	86.3	76.8	86.1	10.6	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.4	140.0	79.4	75.3	134.8	79.4	74.7	131.9	79.7	41.6	85.5	41.0	86.0	40.4	85.8	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.7	17.8	16.4	DB	25.3	107.7	4.3	1.4	22.2	108.7	19.6	109.5	17.2	110.3	14.0	111.2
			WB	23.7	79.1	3.8	1.2	21.0	80.0	18.8	80.7	16.6	81.3	13.8	82.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	74.8	57.1	63.3	69.9	76.8	82.2	87.0	88.5	88.7	83.2	75.5	65.8	58.9	(6)
	DBStd	13.23	9.11	10.37	8.95	6.89	5.56	3.41	2.94	2.99	4.71	7.89	9.54	9.04	(7)
	HDD50	83	36	16	3	0	0	0	0	0	0	0	7	21	(8)
	HDD65	846	277	145	61	7	0	0	0	0	0	16	105	235	(9)
	CDD50	9132	255	388	618	803	998	1111	1193	1201	995	791	482	297	(10)
	CDD65	4421	31	97	212	360	533	661	728	736	545	342	130	46	(11)
	CDH74	53619	291	844	1882	3756	6170	8817	10036	10334	6303	3681	1152	353	(12)
	CDH80	28915	65	301	782	1868	3157	5096	5992	6299	3273	1638	368	76	(13)
(14) Wind	WSAvg	7.3	5.6	6.8	7.8	8.7	9.1	9.2	9.3	8.2	5.9	6.1	5.6	5.2	(14)
(15) Precipitation	PrecAvg	21.3	0.9	0.9	0.9	1.2	2.3	2.0	2.4	1.7	4.2	1.7	1.5	0.8	(15)
	PrecMax	33.3	3.8	3.0	2.6	4.1	6.0	6.1	6.9	3.9	10.3	5.8	3.9	3.3	(16)
	PrecMin	13.3	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.9	0.1	0.0	0.0	(17)
	PrecStd	5.4	1.1	0.9	0.8	1.2	1.7	1.5	2.1	1.3	2.2	1.4	1.2	0.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.0	93.1	96.2	103.4	102.1	106.7	106.0	106.3	102.1	98.0	91.0	86.5	(19)
		MCWB	63.4	64.3	64.9	67.2	72.8	72.7	71.2	71.8	71.5	73.3	70.1	65.5	(20)
	2%	DB	81.3	87.7	91.1	97.4	99.5	102.5	104.0	104.2	99.4	94.2	88.0	81.8	(21)
		MCWB	62.4	65.0	66.3	67.9	73.3	73.1	72.0	72.0	71.8	72.3	69.2	64.9	(22)
	5%	DB	77.0	83.5	88.1	93.5	96.9	100.2	102.2	102.4	97.1	91.3	84.3	78.0	(23)
		MCWB	60.3	65.2	66.8	69.0	72.9	73.1	72.5	72.1	72.2	71.6	68.0	63.7	(24)
	10%	DB	72.6	79.3	84.3	90.3	93.4	98.4	100.1	100.3	94.6	89.2	81.0	73.5	(25)
		MCWB	59.9	64.5	66.2	68.1	72.9	72.9	72.7	72.5	72.2	71.2	66.2	63.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.7	71.0	72.2	75.3	78.2	78.5	78.5	78.4	78.2	77.7	74.9	71.0	(27)
		MCDB	76.7	81.0	80.8	86.4	88.5	90.6	87.5	86.9	86.7	86.0	81.7	76.7	(28)
	2%	WB	67.1	69.4	71.2	73.7	77.1	77.4	77.6	77.5	77.3	76.7	73.2	69.2	(29)
		MCDB	75.1	77.8	80.4	85.2	87.4	87.7	86.7	86.2	84.4	84.3	80.2	75.8	(30)
	5%	WB	65.2	68.5	70.2	72.7	76.1	76.9	76.7	76.8	76.7	75.6	71.3	67.8	(31)
		MCDB	71.7	76.5	79.3	84.1	87.1	87.3	86.3	86.5	84.1	83.6	78.8	74.0	(32)
	10%	WB	63.0	67.4	69.1	71.7	75.2	76.2	76.0	76.1	76.0	74.5	69.6	65.8	(33)
		MCDB	69.1	75.2	78.2	83.0	86.6	86.8	86.0	86.7	83.9	83.1	77.9	72.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	23.6	23.3	22.8	24.9	21.7	22.7	23.3	23.9	20.6	22.7	22.3	21.2	(35)
		MCDBR	28.9	27.7	26.4	27.9	24.7	25.3	26.0	26.3	24.3	22.6	23.7	26.4	(36)
	5% WB	MCWBR	13.8	9.7	8.8	8.2	5.2	4.6	4.7	4.9	5.4	6.1	8.7	12.6	(37)
		MCWBR	20.7	20.7	22.1	23.7	20.5	21.5	21.8	22.0	18.5	18.9	19.2	19.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.326	0.347	0.374	0.398	0.413	0.406	0.399	0.398	0.397	0.364	0.355	0.329	(40)
		taud	2.524	2.461	2.394	2.347	2.347	2.424	2.439	2.465	2.426	2.509	2.472	2.522	(41)
	Ebn at Noon		294	295	291	286	280	279	281	282	280	285	280	287	(42)
		Edh at Noon	29	33	37	40	40	37	36	36	36	31	30	28	(43)
(44) All-Sky Solar Radiation	RadAvg		1022	1255	1548	1802	1919	2184	2162	2094	1699	1455	1132	935	(44)
	RadStd		124	208	196	173	162	97	158	99	143	164	135	143	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (15 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page